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CHAPTER 1 • NUMBER • TOPIC 14

Exchange Rates & Best Buys

Multiply or divide by the rate — the side with 1 tells you which. Then compare like with like: same currency, same unit.



$$\frac{3}{10} = 0.3 =$$

30%**3**

VARIANTS

1

PAST-PAPER PART

32

PRACTICE QUESTIONS

YOU WILL MASTER**V1** Converting currencies**V2** Compare in two currencies**V3** Best buys

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Money at a glance



Your notes

Exchange Rates & Best Buys

CONVERT

V1 Converting currencies

$$\$306 \div 1.125 = €272$$

PRACTICE ONLY • BE READY

COMPARE COUNTRIES

V2 Compare in two currencies

Dane vs Arctic Oil

PAST PAPERS: 1 OF 1

COMPARE SIZES

V3 Best buys

$$\frac{650}{30} = 21.7\text{p per plant}$$

PRACTICE ONLY • BE READY

THE KEY QUESTION

WHICH SIDE IS 1?

£1 = €1.16: from £ you
×1.16; to £ you ÷1.16.

THE KEY HABIT

LIKE WITH LIKE

Same currency and same
unit before you compare.

★ How often does each variant come up?

1 past-paper question (Jan 2020 P2H, 4 marks); 32 practice parts — grouped by type. Currency and best value also turn up inside percentage questions.

	PAST PAPERS (OUT OF 1)		PRACTICE PARTS
V1 Converting currencies	0		11
V2 Compare in two currencies	1		9
V3 Best buys	0		12

WHAT THIS TELLS YOU Rare on its own, but when it comes it is a 4-mark “show your working” comparison. One clear unit price for each option, then a sentence, gets every mark.

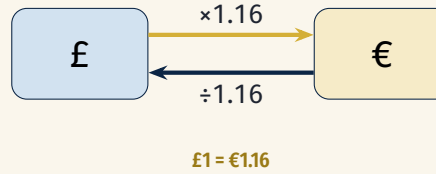
BEFORE YOU START

The big idea

**An exchange rate is a multiplier.**

£1 = €1.16 means every pound becomes 1.16 euros. Going **away from the 1**: multiply. Going **towards the 1**: divide.

A best buy is the same idea:
price per one unit.


Two tools
From the 1

£200 → ×25.82
£ into koruna:
multiply.

To the 1

€5 ÷ 1.16
Euros into £:
divide.

Unit price

$\frac{\text{price}}{\text{amount}}$
Pence per gram —
smaller is better.

Amount per £

$\frac{\text{amount}}{\text{price}}$
Grams per penny
— bigger is better.

Sense check

£1 buys **more** than 1 euro
so euros > pounds

€5 should be **less** than £5
 $5 \div 1.16 = £4.31$ ✓

wrong way gives £5.80
— bigger, so ×

**CALCULATOR FIRST, THEN CHECK**

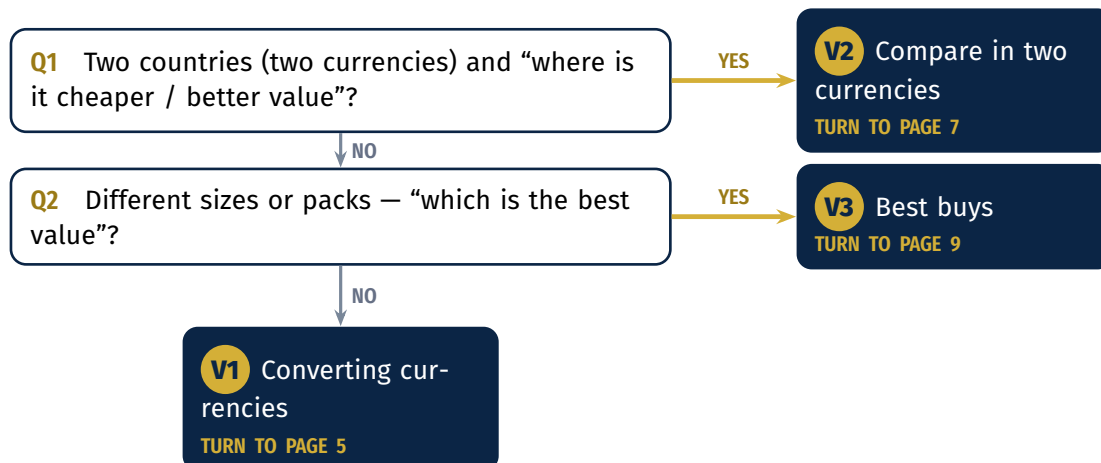
Calculator first, then check: does the answer go the right way (more euros than pounds)?
Money answers to 2 d.p.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD



What is it?



When you see → you do



How do I do it?



Watch out



Worked example



Exam tip



Check yourself



Calculator check



VARIANT 1

Converting Currencies

Away from the 1: multiply. Towards the 1: divide.

SEEN IN PAST PAPERS

● 0/1

NOT ASKED YET • BE READY

PRACTICE QUESTIONS: 11



Your notes

What is it?

In plain words

Look at the rate: which currency has the 1?

Changing **out of** that currency:
multiply by the rate. Changing **into**
it: divide. Round money to 2 d.p.;
notes round **down**.

€1 = \$1.125

\$306 $\rightarrow$ € (towards the 1)

$$306 \div 1.125 = \text{€}272$$

IT LOOKS LIKE

€1 = \$1.125

\$1 = 68.14 rupees

100-koruna notes

How do I do it?

1 Find the 1

Which currency has 1 in the rate?

2 Direction

Out of it: $\times$ rate. Into it: $\div$ rate.

3 Calculate

Full display.

4 Round

Money 2 d.p.; whole notes round down.

WHICH SIDE IS 1? It decides $\times$ or $\div$.

V1 Converting Currencies

VARIANT 1 • SEE IT DONE, THEN TRY

 Your notes

WHEN YOU SEE → YOU DO

"Exchange rate is €1 = \$1.125 ... change \$306 into euros" → **Into the 1:** $\div 1.125$

"Changed \$300 into 3891 Namibian dollars. \$1 = ..." → **Rate** = $3891 \div 300$

"As many 100 koruna notes as possible" → **Convert, then round down to whole 100s**

WORKED EXAMPLE PRACTICE

Practice • Easy • Q10 • 2 marks

Adele changes \$306 into euros (€) when the exchange rate is €1 = \$1.125. Calculate the number of euros she receives.

STEP 1 €1 = \$1.125 ◀ the 1 is on the euro side

STEP 2 \$ → € : towards the 1 ⇒ $\div$ ◀ direction

STEP 3 $306 \div 1.125 = 272$ ◀ calculate

ANSWER €272

✓ $272 \times 1.125 = 306$ ✓

WATCH OUT

✗ $306 \times 1.125 = 344.25$

✓ $306 \div 1.125 = 272$

Each euro costs more than a dollar, so you get fewer euros.

EXAM TIP

Undo it as a check: convert your answer back and you should get the start.

Check yourself

calculator first • answers on page 11

1.1 Practice • Easy

Rashid changes 30 000 rupees to dollars; \$1 = 68.14 rupees. How many dollars?

Answer:

1.2 Practice • Easy

Karel changed \$300 into 3891 Namibian dollars. \$1 = ... Namibian dollars.

Answer:

1.3 Practice • Medium

Up to £200 into 100-koruna notes; £1 = 25.82 koruna. How many notes?

Answer:

V2**VARIANT 2** MOST COMMON IN THE EXAM

SEEN IN PAST PAPERS

● 1/1

PRACTICE QUESTIONS: 9

Compare in Two Currencies

Convert one price so both are in the same currency (and per the same unit), then compare.

Came up in: Jan 2020 P2H Q12


Your notes

? What is it?

In plain words

Prices in two countries can't be compared until they are in **one currency**. Convert one of them. If the amounts differ too (litres, kg), find the **price per unit** as well. Then say which is cheaper and by how much.

IT LOOKS LIKE

HK\$3179.55 vs £285

krone vs dollars per litre



How do I do it?

1 Currency

Convert one price into the other currency.

2 Unit

Different amounts? Price ÷ amount for each.

3 Compare

Same currency, same unit.

4 Conclude

Name the cheaper one; give the difference if asked.

LIKE WITH LIKE Same currency, same unit.

V2 Compare in Two Currencies

VARIANT 2 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

"Where is it cheaper? How much cheaper?"

→ **Convert, compare, subtract**

Different quantities (4.2×10^5 litres, 8.6×10^5 litres)

→ **Price per litre for both**

"You must show your working"

→ **Two clear values + a sentence****WORKED EXAMPLE PAST PAPER**

January 2020 · Paper 2H · Q12 · 4 marks

Dane Oil: 4.2×10^5 litres for 2 500 000 krone. Arctic Oil: 8.6×10^5 litres for 770 000 dollars. 1 dollar = 6.57 krone. Which is better value?

STEP 1 $2\,500\,000 \div 6.57 = \$380\,517.50$ ◀ Dane into dollars

STEP 2 $380\,517.50 \div (4.2 \times 10^5) = \0.906 ◀ Dane per litre

STEP 3 $770\,000 \div (8.6 \times 10^5) = \0.895 ◀ Arctic per litre

STEP 4 $0.895 < 0.906$ ◀ compare

ANSWER Arctic Oil (\$0.895 per litre) ✓ both in dollars per litre ✓

WATCH OUT

× $2\,500\,000 > 770\,000$, so Arctic
 ✓ compare \$ per litre

Different currencies and amounts: the raw prices mean nothing.

EXAM TIP

Keep 3 s.f. or more for the per-litre prices — 0.906 and 0.895 are close.

Check yourself

calculator first · answers on page 11

2.1 Practice · Medium

Camera: HK\$3179.55 in Hong Kong, £285 in London. £1 = HK\$12.30. How much cheaper in Hong Kong?

Answer:

2.2 Practice · Medium

Newspaper: 5 euros in France, £2.50 in England. £1 = €1.16. Difference?

Answer:

2.3 Practice · Medium

Wallet: 31 euros in France, £23.50 in London. £1 = €1.34. How much cheaper in France?

Answer:



VARIANT 3 Best Buys

Price per one unit for every option; the smallest wins.

SEEN IN PAST PAPERS

● 0/1

NOT ASKED YET • BE READY
PRACTICE QUESTIONS: 12



Your notes

What is it?

In plain words

Packs of different sizes: work out the **price of one** (one plant, one gram, one pint) for each.

The **lowest unit price** is the best value. Work in pence to avoid tiny decimals.

30 for £6.50	40 for £8.95	50 for £10.99
21.7p	22.4p	22.0p

SMALLEST PRICE PER PLANT WINS

IT LOOKS LIKE

30 for £6.50	40 for £8.95
50 for £10.99	

How do I do it?

- | | |
|-------------------|-------------------------------------|
| 1 Pence | Change £ to pence. |
| 2 Divide | Price ÷ amount for each option. |
| 3 Compare | Same unit, enough decimal places. |
| 4 Conclude | Name the best option in a sentence. |

PRICE ÷ AMOUNT Smallest unit price is best value.

V3 Best Buys

VARIANT 3 · SEE IT DONE, THEN TRY


 Your notes
WHEN YOU SEE → YOU DO

“Which is the best value for money?” → **Unit price for every option**

Prices in £, amounts in grams or ml → **Pence per gram / per ml**

“You must show all your working” → **Every unit price written + a conclusion**

WORKED EXAMPLE **PRACTICE**

Practice · Easy · Q1 · 4 marks

A small tray of 30 plants costs £6.50, a medium tray of 40 costs £8.95, a large tray of 50 costs £10.99. Which tray is the best value?

STEP 1 650, 895, 1099 pence ◀ into pence

STEP 2 $650 \div 30 = 21.7\text{p}$, $895 \div 40 = 22.4\text{p}$ ◀ per plant

STEP 3 $1099 \div 50 = 22.0\text{p}$ ◀ per plant

STEP 4 $21.7 < 22.0 < 22.4$ ◀ compare

ANSWER the small tray

✓ lowest price per plant ✓

WATCH OUT

✗ “Large is always best value”

✓ work out each unit price

Here the small tray wins.

EXAM TIP

Write the conclusion with the numbers:
 “small: 21.7p per plant, the lowest”.

Check yourself

calculator first · answers on page 11

3.1 Practice · Easy

Blueberries: 125 g for £1.60 or 225 g for £2.80. Better value?

Answer:

3.2 Practice · Medium

Milk: 4 pints for £1.18 or 6 pints for £1.74. Better value?

Answer:

3.3 Practice · Medium

Potatoes: 12.5 kg for £9 (farm) or 2.5 kg for £1.83 (supermarket). Better value?

Answer:

CHECK YOURSELF

Answers



Your notes

Every answer below was checked in Python (check_answers.py).

1.1 \$440.27**1.2** 12.97**1.3** 51 notes**2.1** £26.50**2.2** £1.81**2.3** 37p**3.1** large carton**3.2** 6 pints**3.3** farm shop

THE WHOLE TOPIC ON ONE PAGE

Money cheat sheet



Your notes
V1 Converting

$$\$306 \div 1.125 = \text{€}272$$

Away from the 1: $\times$; towards it: $\div$.

V2 Two currencies 1/1

$$\$0.895 < \$0.906 \text{ per litre}$$

Same currency, same unit, then compare.

V3 Best buys

$$650 \div 30 = 21.7\text{p}$$

Price $\div$ amount; smallest wins.

**TOP 3 LOST MARKS**

1. $\times$ instead of $\div$ $\times$
2. comparing different currencies $\times$
3. no conclusion sentence $\times$

BOTTOM LINE Which side is 1? • like with like • price $\div$ amount • say which, with numbers.